

Topic: Join Operation

Q. Explain the Join operation in Relational Algebra. Describe the different types of Joins including Theta Join, Equi Join, Natural Join, and the Outer Joins.

****Definition to Remember****

A **Join
 fundam
 operatio

Join Operations (join)

(Return only rows with a matching counterpart in both tables)

A. Theta Join (join_{})

* Joins

B. Equi Join

* A **S

C. Natural Join (join)

* An **

2. Types of Outer Joins

(Preserve unmatched rows - missing values are filled with NULL)

STUDENT ENROLLMENT

| Join Type | What is Returned |

| :--- | :-

Example (Left Outer Join): STUDENT Left Join ENROLLMENT -> Shows ALL students. Luke has no enrollment -> his Course column = NULL.

3. Summary: All Join Types

| Join Type | Returns | Duplicates |

| :--- | :-

*****Must-Write Points (for 10 marks)*****

1. Join =

*****Quick Recall*****

`Join =
Left/Rig